

Lab 3, Act 1 – Answer Key

Trusting a positive detection

NRES 746 – Instructor Answer Key (not for student distribution)

This key follows the same section structure as the student handout. Questions with a single correct numeric or derived answer are answered directly; open-ended items (interpretation, discussion) are given sample answers/discussion points rather than a fixed “correct” response – use these to gauge whether a student’s reasoning is on track, not as an exact-match rubric. Act 1 is deliberately discovery-based (no formula is given up front), so expect more variation in *how* students get to a correct answer than in later Acts.

Part 1: Count your way to an answer

1a. Completed table ($N = 100,000$), rounded to the nearest whole number:

```
N <- 1e5
species      <- N * p_S
not_species  <- N * (1 - p_S)
flag_S       <- species * sens
flag_notS    <- not_species * fpr
data.frame(
  row        = c("Actually the species", "Not the species", "Total"),
  flagged    = round(c(flag_S, flag_notS, flag_S + flag_notS)),
  not_flagged = round(c(species - flag_S, not_species - flag_notS,
                        (species - flag_S) + (not_species - flag_notS))),
  total      = c(species, not_species, N)
)
```

```
##           row flagged not_flagged total
## 1 Actually the species      10         0     10
## 2      Not the species    1000     98990 99990
## 3              Total    1010     98990 100000
```

1b. $PPV = 10/1010 \approx 0.0099 \approx 1\%$ – pure counting, no formula needed.

1c. No fixed answer, but a correct response should register that $\approx 1\%$ is *surprisingly low* given how good 98% sensitivity / 1% false-positive rate sounded at the start. A good response to the field team: a single flagged image alone isn’t enough to act on confidently – look for corroborating evidence (a second independent detection, a track, scat) before committing resources to that flag.

Part 2: Turn your counting into a general recipe

2a. Flagged-and-truly-the-species count = $N \times \text{prevalence} \times \text{sensitivity}$. Total flagged count = $N \times \text{prevalence} \times \text{sensitivity} + N \times (1 - \text{prevalence}) \times \text{false positive rate}$.

2b. N cancels from top and bottom, leaving:

```
ppv <- (p_S * sens) / (p_S * sens + (1 - p_S) * fpr)
ppv
```

```
## [1] 0.009705853
```

$$\text{PPV} = \frac{\text{prevalence} \times \text{sensitivity}}{\text{prevalence} \times \text{sensitivity} + (1 - \text{prevalence}) \times \text{false positive rate}} \approx 0.0097$$

2c. Any phrasing that captures the formula above is correct – word-for-word notation doesn’t matter. This is Bayes’ rule/PPV, arrived at without a name; don’t volunteer the name “Bayes’ rule” when grading this – that reveal is Wednesday’s lecture.

Common snags: students who divide by $P(S)$ (the marginal probability of being the species) instead of $P(F)$ (the marginal probability of being flagged) in 2b – this produces a PPV > 0.5 at Site A, which is the tell. Students who try to cancel something other than N in 2b, or who stall because they’re trying to recall “the formula” instead of generalizing their own Part 1 arithmetic.

Part 3: Try it somewhere else

3a. Site B (prevalence = 0.01, same sensitivity/false-positive rate):

```
p_S_B <- 0.01
ppv_B <- (p_S_B * sens) / (p_S_B * sens + (1 - p_S_B) * fpr)
ppv_B
```

```
## [1] 0.4974619
```

PPV $\approx 0.497 \approx 50\%$.

3b. No fixed answer, but a strong response identifies that the classifier’s own properties (sensitivity, false-positive rate) were *identical* at both sites – only prevalence changed, and yet PPV swung from $\approx 1\%$ to $\approx 50\%$. The intended takeaway: prevalence (the “base rate”) matters at least as much as the classifier’s own accuracy, and a classifier that sounds great in isolation can still be dominated by false positives when the target is rare. Students who bring up the “base rate fallacy” by name (even loosely) should get full credit for connecting today’s exercise to that idea – the formal term arrives Wednesday.

Group discussion and revision

No fixed answer – the point is for students to compare their Part 2 recipes and reconcile any differences in notation or approach, and to flag their own muddiest points.